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Nachiket Mhatre

feedback@digit.in

Informally known as the Shinshin (spirit of the heart) in Japan, the Mitsubishi ATD-X is the country's first indigenously designed and developed fifth-generation air superiority fighter. In fact, by the time this issue is printed, the jet will have embarked upon its maiden flight if all goes according to plan. Does that mean the Shinshin stealth fighter will beat USAF's long delayed F-35 Lightning II as the second ever fifth-generation jet fighter to enter active service since the Lockheed Martin F-22 Raptor? Not even by a long shot.

More concept than reality

The Shinshin's military designation is ATD-X for a reason, as it stands for Advanced Technology Demonstrator. Even in its latest X2 iteration, which is poised to take first flight shortly, the Shinshin can't even be called a prototype,

but a proof-of-concept of sorts whose main purpose is to serve as a platform for all the subsequent research and development required to come up with a viable fifth-generation air superiority fighter design. The Shinshin only serves as a stepping stone to the real frontline stealth jet fighter designated as the F-3, which won't be going into production any time before 2027.

To put this into perspective, Lockheed Martin's own fifth-generation stealth fighter jet, the F-35 Lightning II, has been in production since 2006, but it has yet to be inducted into service. Needless to say, don't get too excited because it will be decades before the Shinshin joins active service; if at all, that is, because let's not forget the X2 is nothing more than a proof-of-concept, and not even a prototype. It isn't even the only fifth-generation fighter aircraft under development. A whole host of nations from the minnows Iran and Turkey to the formidable China, Russia, and even India (with Russian assistance)

are working on their own answers to the American F-22 and F-35. The Russian Sukhoi Pak FA and the Chinese Shenyang J-31 have already completed their first flights and the former is slated to be rolled out into service in 2017.

The blueprints

However, before we get into the nitty gritty of the Japanese fifth-generation jet fighter, it's prudent to understand what the term entails, and why Japan has been pursuing it aggressively. Military jet fighters are classified according to the sophistication of their avionics (aircraft electronics), radars, communication, armament, and flight characteristics such as manoeuvrability and stealth technologies. To give a brief primer on the important fighter generations, the third-generation aircraft focused on development of sophisticated attack capabilities by employing missiles and smart bombs that relied on radar, thermal, and laser guidance.

The fourth-generation jet fighters deviated from the last generation's overde-